AeroCast-AI

Day-Ahead Flight Delay Prediction

Predicting delay risk before the next day's flights take off.



Our project focuses on predicting which flights are likely to arrive late, using information available the evening before departure. In Phase 1, we built and compared machine learning models for this task.

TEAM MEMBERS **T-049**

Nipun Patlori	2401010323
Mohan Kumar C R	2401010282
Anushka Tyagi	2401010090
Omkar Hadole	2401010175
Krish Dabas	2401010233

Problem Statement & Our Approach

“Can we predict tomorrow's flight delays today?”

The Problem

Flight delays disrupt schedules and can affect many later flights. Airlines need an early warning to plan ahead.

Our Approach

Use historical flight records and machine learning to estimate the risk of a flight arriving 15 minutes or more late.

Our prediction deadline

8:00 PM · Previous day

(Only information available by this time can be used)

Delayed Flight — the one arriving **15 minutes*** or more after its scheduled arrival time

Phase 1 Goal — Identify high-risk flights early and compare two machine learning approaches fairly.

**This 15-minute cutoff follows the US Bureau of Transportation Statistics' on-time arrival rule.*

Understanding Dataset

336,776	2013	3	≥ 15 min
Original Flight Records	Year	Origin Airports (EWR, JFK, LGA)	Delay Threshold

Binary Target Class Definition: Target variable → (is_delayed)

✓ Class 0: NOT LATE

Arrival delay < 15 minutes

! Class 1: LATE

Arrival delay ≥ 15 minutes

Previous day, 8 PM → Model predicts → Next day's arrival

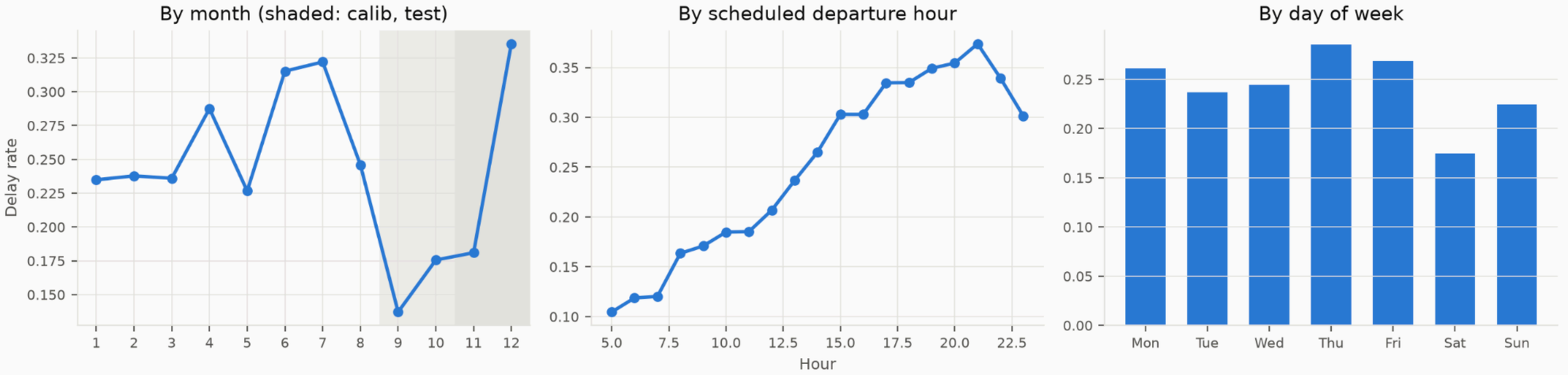
1. **Supervised Learning**
2. **Binary Classfication**
3. **Time-dependent prediction**

Chronological split (train on the past, test on the future; no shuffling)

TRAIN	CALIBRATION	TEST
8 Jan – 31 Aug 2013	Sep – Oct	Nov – Dec
211,684 flights · 26.5% late	**55,628 · 15.7% late**	**53,991 · 25.8% late**

Cancelled (8,255) and diverted (1,175) flights have no arrival delay, so they are left out of the target but still counted in schedule congestion. Overall, 24.5% of flights are late.

Key Delay Patterns (EDA)



Season

Dec, Jul and **Jun** are worst (over **30%**); **Sep** to **Nov** are calmest (under **20%**)

Hour of Day

About **10%** of 5 am flights are late, rising to about **37%** at 9 pm

Day of Week

Saturday is the best day (about 18% delayed)

Feature Engineering & Leakage Prevention

43 features engineered across **5** categories.

1. CATEGORICAL

2. SCHEDULE

3. CONGESTION

4. ENTITY HISTORY

5. DAY-LEVEL HISTORY

Three automated leakage checks ran; one detected two same-day aircraft-rotation features that violated the cutoff constraint. They were removed.

45 → 43 features

Training & Model Selection



Models We Trained

- **Logistic Regression: L2-regularised (C = 0.1), missing values median-imputed with flags.**
- **XGBoost: shallow trees (depth 2, 108 trees), handles missing values natively.**
- **Baselines: base-rate prior and route-history delay rate.**

Model Selection

- **Chronological splits:** train Jan–Aug, calibrate Sep–Oct, test Nov–Dec.
- **Rolling-origin CV:** 4 folds validate May, Jun, Jul and Aug, each trained only on earlier months.
- **Rule fixed before test:** pick the higher mean CV PR-AUC.

CV PR-AUC Results

Base-rate prior
0.265
XGBoost
0.450
Logistic Regression ✓
0.471

Why We Chose Logistic Regression

Our final model is **Logistic Regression**. Four simple reasons:

1

We fixed the rule before testing

Pick the model with the higher average CV PR-AUC. LogReg scored 0.471 vs 0.450 and won 3 of 4 months.

2

XGBoost's test lead is not real proof

It is ahead by only 0.014, and the 95% range [−0.010, +0.040] includes zero. That gap could be luck.

3

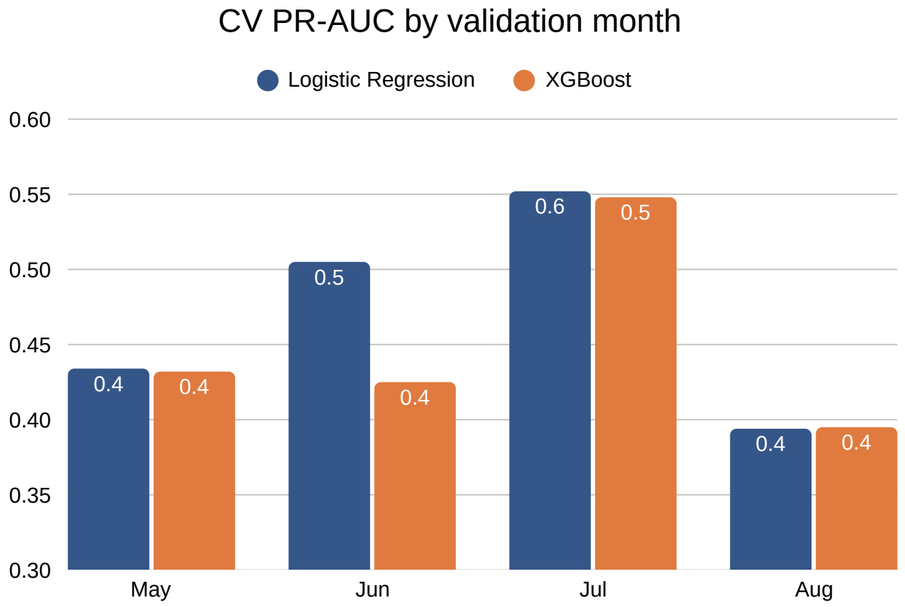
Switching now would be unfair

Changing the model after seeing test scores means choosing by the test set, which spoils the evaluation.

4

Simpler and easier to explain

One coefficient per feature that we can read and defend. With equal performance, we keep the simpler model.

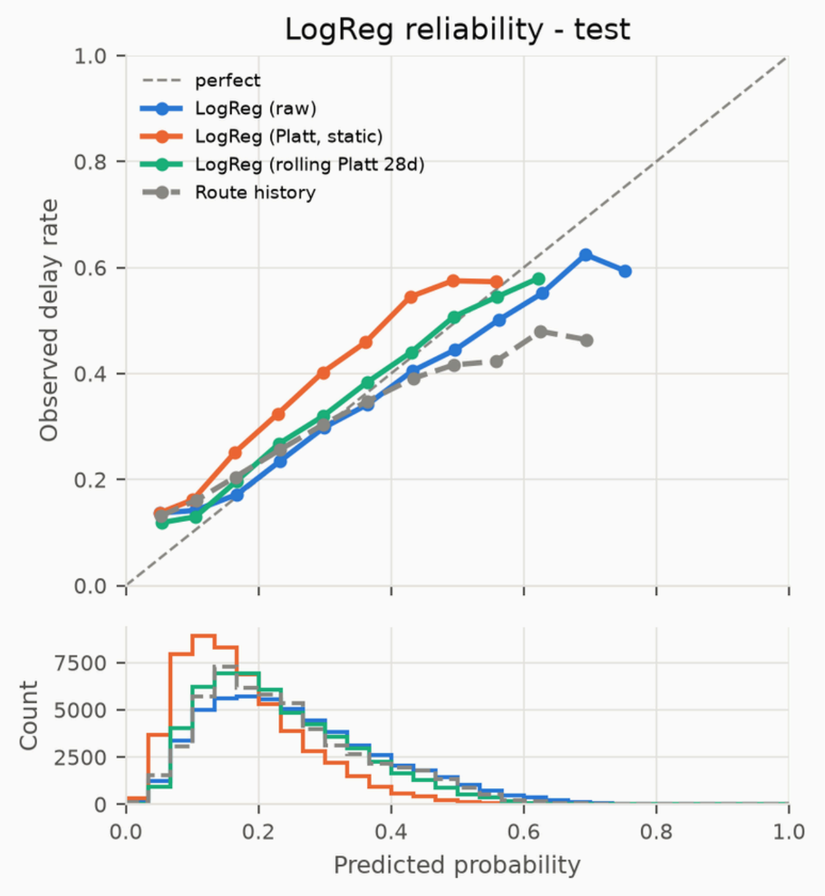


Axis starts at 0.3 · Mean: LogReg 0.471, XGBoost 0.450

In one line

Both models perform about the same, so we keep the one our pre-planned rule picked: the simpler LogReg.

Calibration & Operating Policies



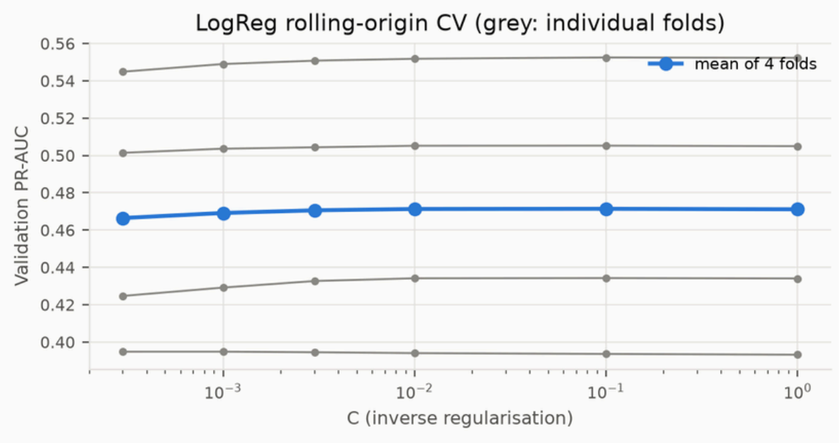
Platt calibration

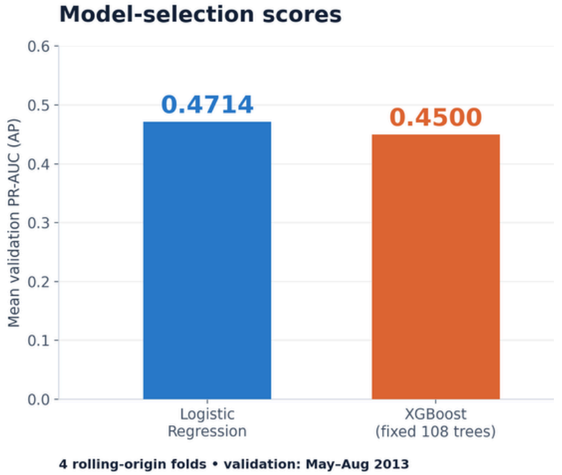
Fit on September, checked on October, frozen before the first November prediction.
Test ECE: 0.027 (rolling) vs 0.019 (raw)

Frozen threshold

0.507908

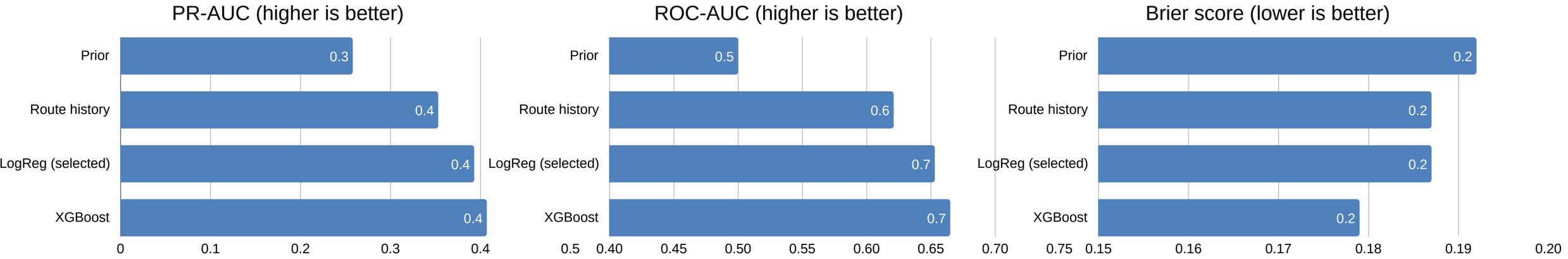
Chosen on Sep–Oct labels to target precision ≥ 0.40, then locked before test.





Test Results & Model Comparison

Test set: Nov–Dec 2013 · 53,991 flights · base rate 25.8%



Both models beat the baselines

LogReg **0.393** and XGBoost **0.407** PR-AUC, both well above route history (0.353) and the base rate (0.258).

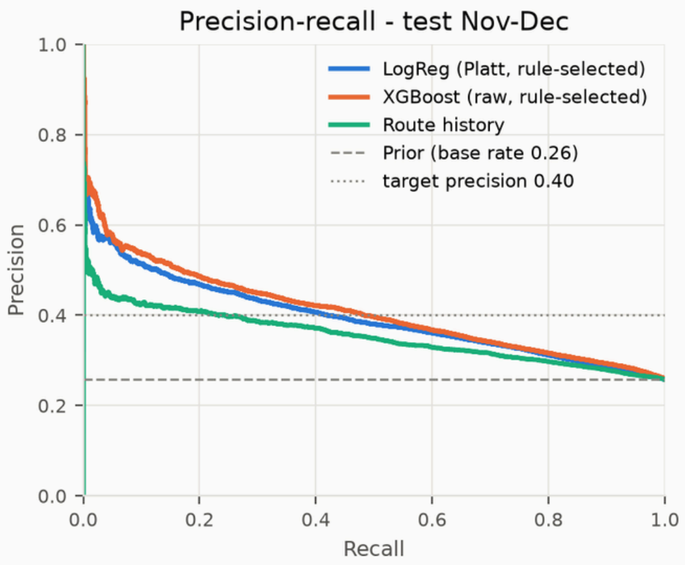
The gap is too small to trust

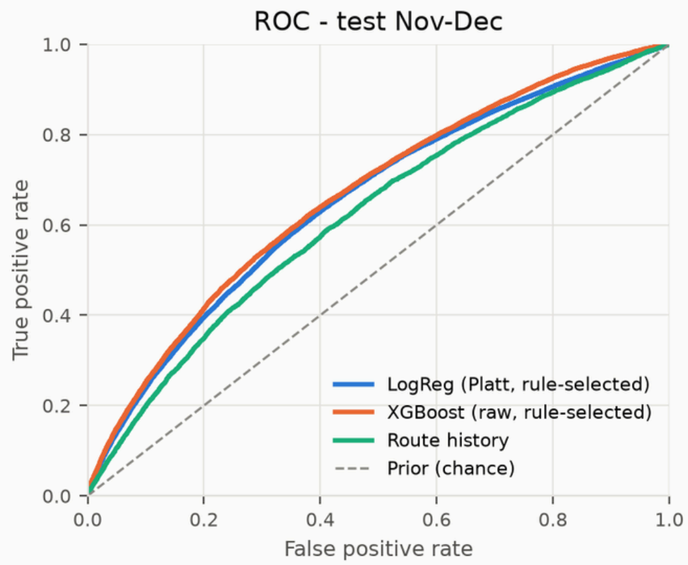
XGBoost is ahead by only **0.014**. The 95% bootstrap range **[−0.010, +0.040]** includes zero, so it could be luck.

No conclusive test winner

The test set cannot tell the two models apart. **We keep Logistic Regression (next slide).**

What We Found





Data	Flights	Really delayed	Real delay %	Model predicted (raw)	After calibrati
Training (Jan–Aug)	211,684	56,074	**26.5%**	26.5%	not applied
Calibration (Sep–Oct)	55,628	8,731	**15.7%**	23.0% ❌ too high	**15.6%** ✅
Test (Nov–Dec)	53,991	13,946	**25.8%**	25.8%	23.2%

Both models beat the baselines

Their curves sit above route history and the base rate across the PR and ROC plots.

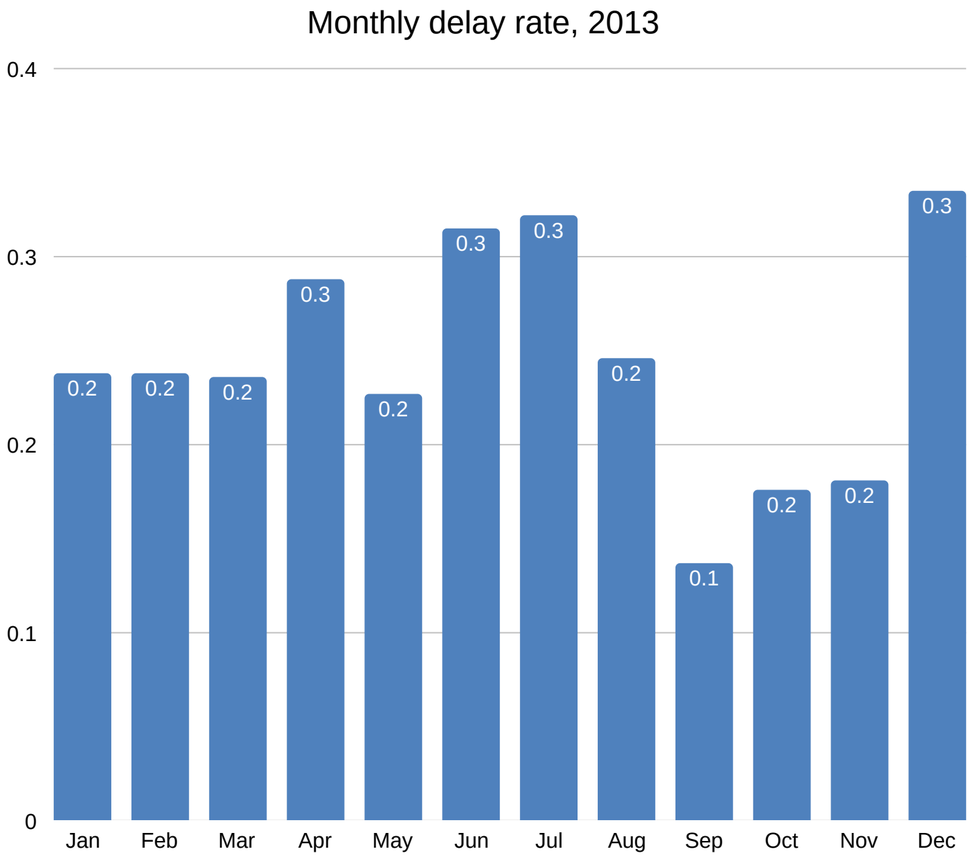
LogReg and XGBoost almost overlap

Their curves are nearly the same, which is why the test set gives no clear winner.

Useful for daily planning

Reviewing the top 10% riskiest flights each day: 44.8% are truly late

Limitations



Narrow scope

One year (2013) and three NYC airports (EWR, JFK, LGA): cross-year and cross-airport generalisation is not shown.

Changing delay rates

13.7% in September vs 33.5% in December. The static calibrator fails in December (ECE 0.143).

Evaluation limits

Only 4 CV folds and one 2-month test window. No weather and no aircraft-rotation features.

Conclusion & Future Work

Conclusion

We can flag risky flights a day ahead using only schedules and past outcomes, with no leakage. Logistic Regression is our final model: as good as XGBoost here, and simpler to explain.

Future work

1. **Deep learning: an MLP with embeddings**
   Phase 2 DL model: learn airline, origin and destination embeddings to capture interactions our hand-made features miss.

2. **What-if rescheduling simulation**
   Try feasible alternative schedules for high-risk flights and estimate how much delay they would save.

References

[1] S. Kaufman, S. Rosset, C. Perlich, O. Stitelman, “Leakage in Data Mining: Formulation, Detection, and Avoidance,” ACM Trans. Knowledge Discovery from Data, vol. 6, no. 4, 2012. https://doi.org/10.1145/2382577.2382579

[2] C. Guo, G. Pleiss, Y. Sun, K. Q. Weinberger, “On Calibration of Modern Neural Networks,” Proc. 34th ICML, pp. 1321–1330, 2017. https://proceedings.mlr.press/v70/guo17a.html

[3] Kaggle Flights dataset (derived from the nycflights13 R package). https://www.kaggle.com/datasets/mahoora00135/flights

[4] Team T49, AeroCast-AI code, notebooks and results. https://github.com/nipun1803/AeroCast-AI